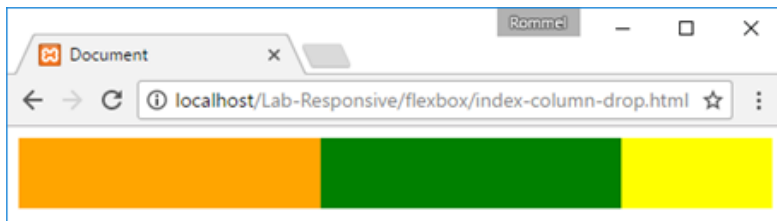
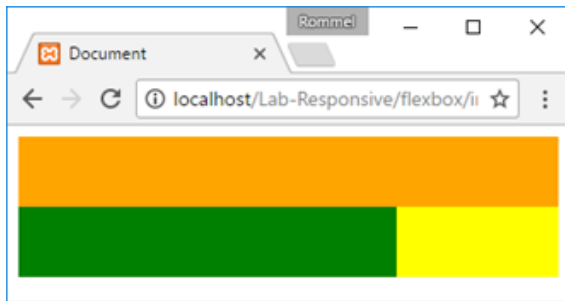
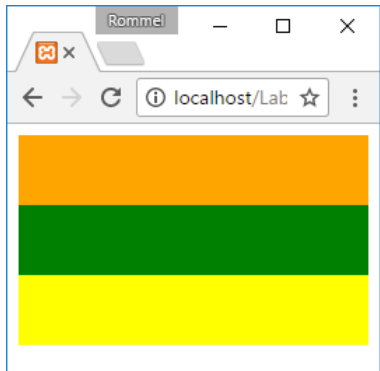


Layouts Flex e Grid

Prof. Rommel Carneiro  

Layout Flexbox



```
<style type="text/css">
.container { display: flex; flex-wrap: wrap; }
div { height: 80px; width: 100% }

.orange { background-color: orange; order: 1; }
.green  { background-color: green;  order: 2; }
.yellow { background-color: yellow; order: 3; }

@media screen and (min-width: 400px) {
  .orange { width: 100%; }
  .green  { width: 70%; }
  .yellow { width: 30%; }
}

@media screen and (min-width: 550px) {
  .orange { width: 40%; }
  .green  { width: 40%; }
  .yellow { width: 20%; }
}
</style>
```

Exercício de Responsividade

Cabeçalho

Notícia

Notícia

Notícia

Notícia

Cabeçalho

Notícia

Notícia

Notícia

Notícia

Cabeçalho

Notícia

Notícia

Notícia

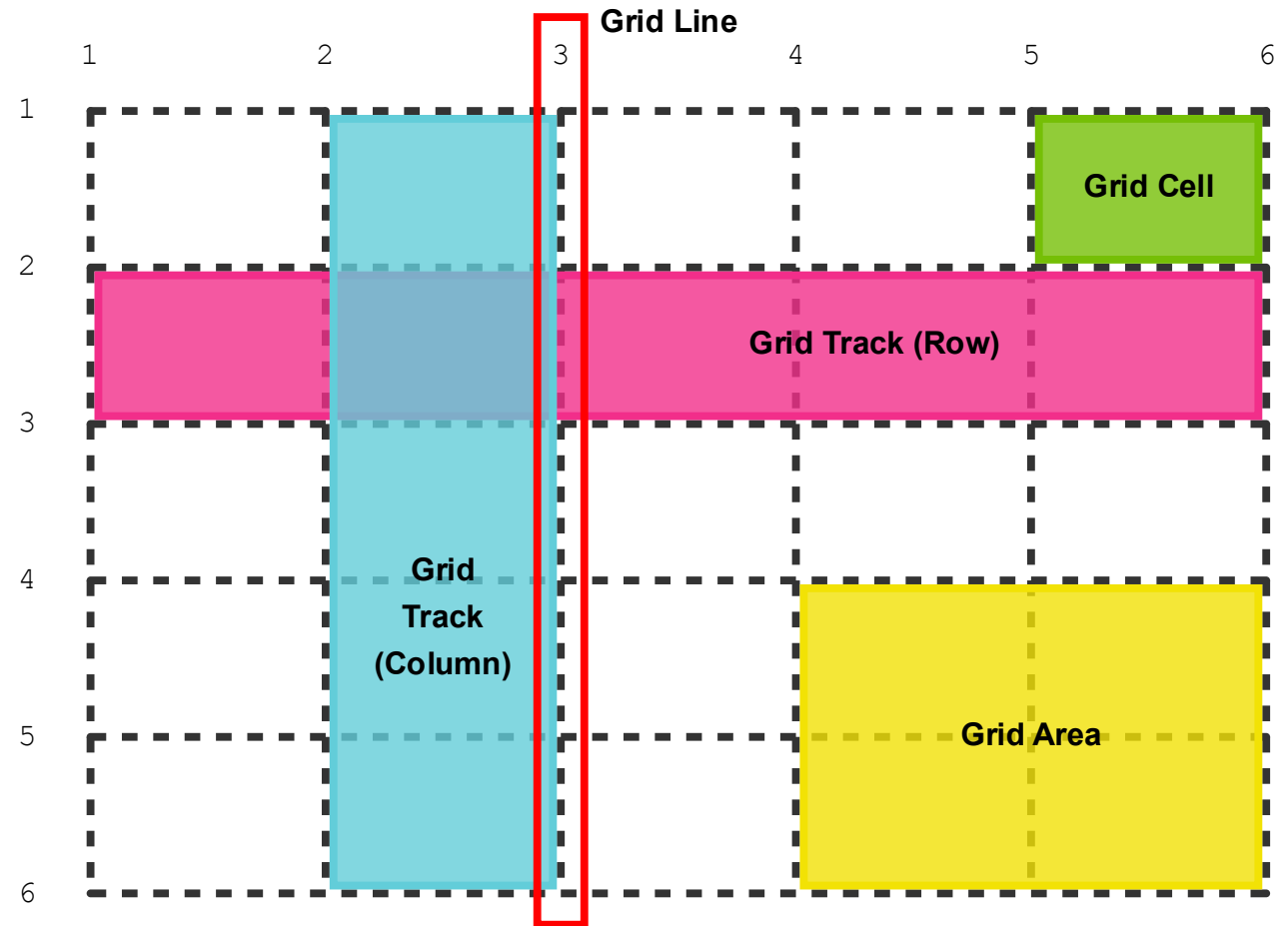
Notícia



Layout Grid

Conceitos fundamentais

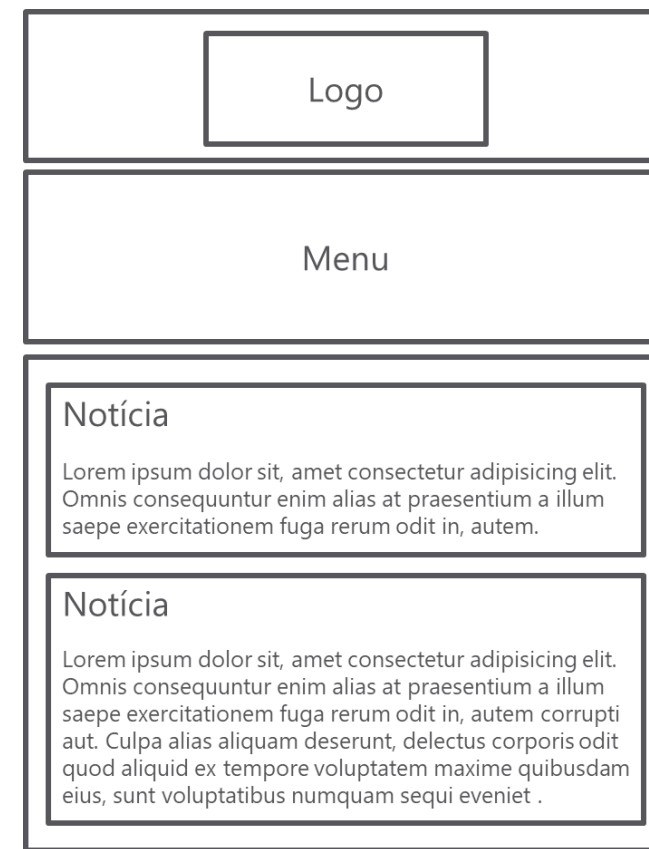
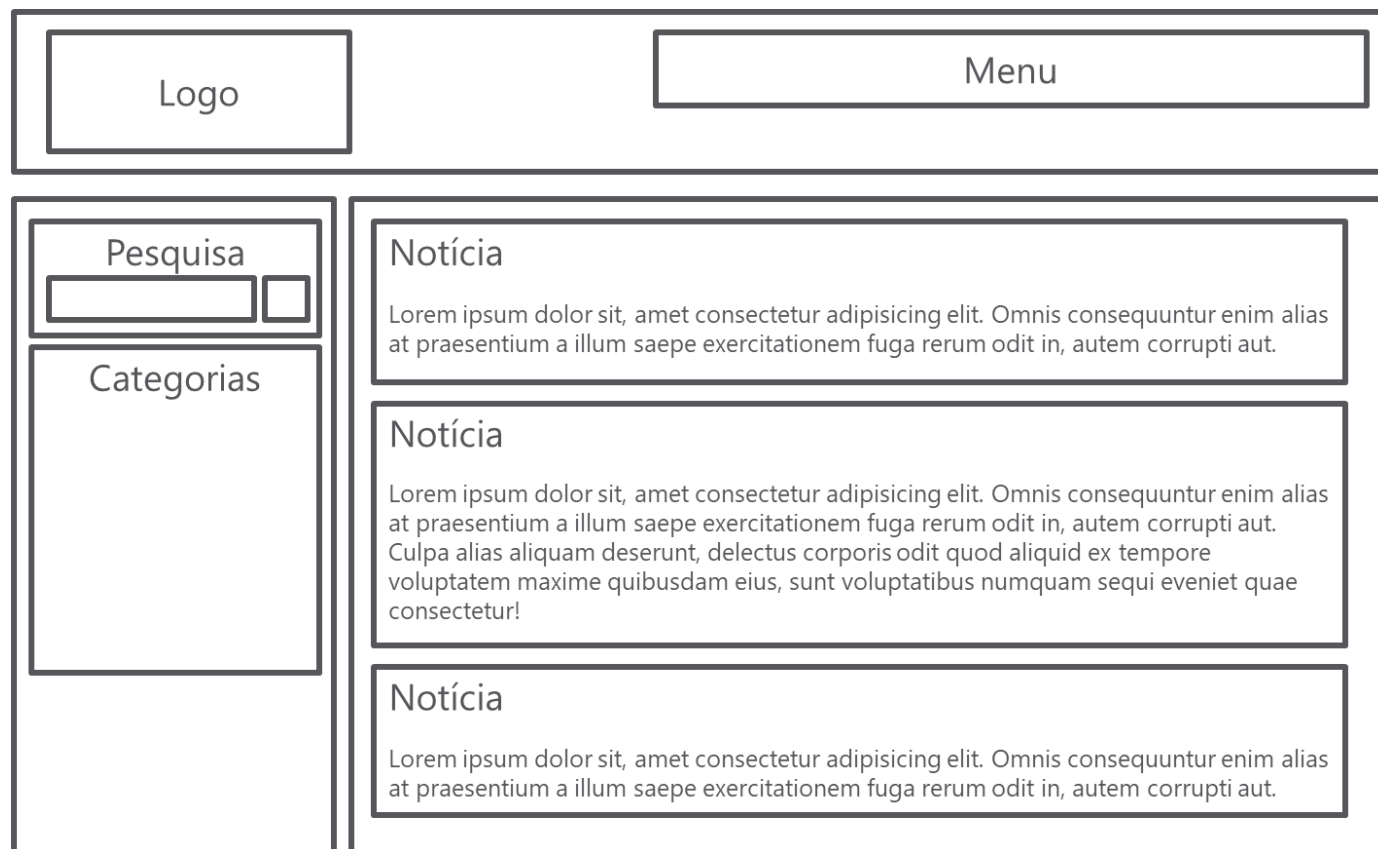
- Lines (verticais e horizontais)
- Tracks (linhas ou colunas)
- Cells
- Areas



Fontes:

- A Complete Guide to Grid - <https://css-tricks.com/snippets/css/complete-guide-grid/>
- CSS Grid Garden - <http://cssgridgarden.com/>
- CSS Grid Layout - <https://webkit.org/blog/7434/css-grid-layout-a-new-layout-module-for-the-web/>

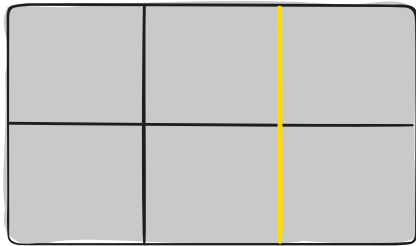
Exercício de Responsividade



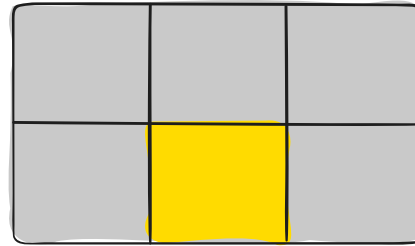
Layout Grid

* CSS-TRICKS

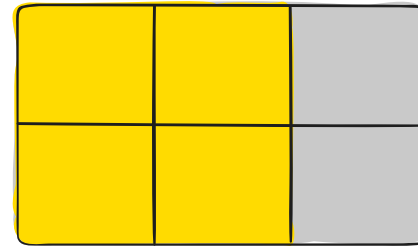
Conceitos fundamentais



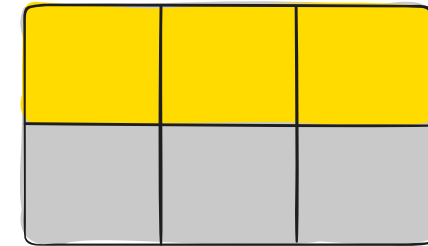
Line



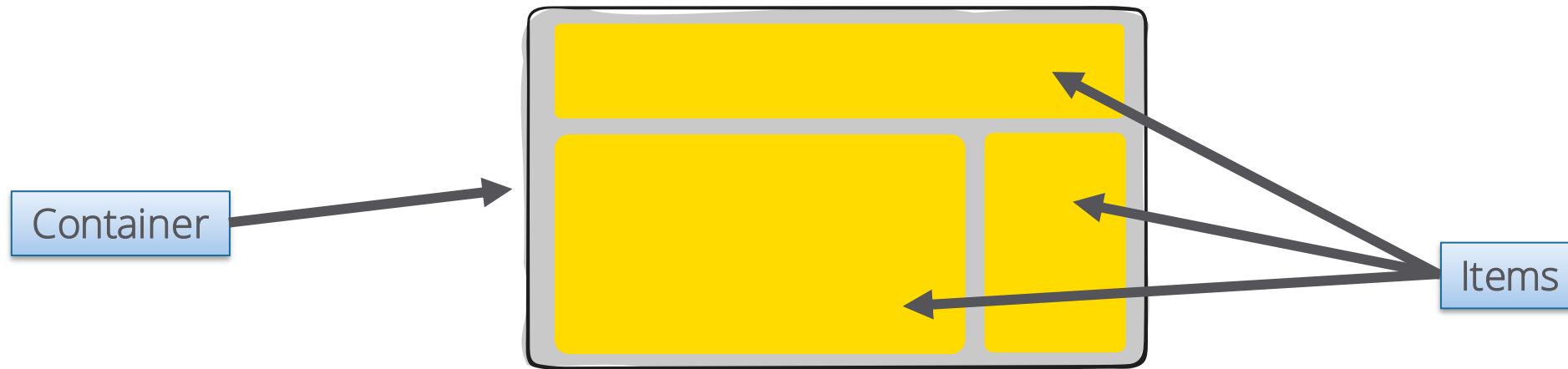
Cell



Area



Track
(linha ou coluna)



Layout Grid

* CSS-TRICKS

Propriedades de Containers

- display: **grid** | **inline-grid** | **subgrid**
 - grid-template-columns: <track-size> ... | <line-name> <track-size> ...
 - grid-template-rows: <track-size> ... | <line-name> <track-size> ...
 - grid-template-areas:
 - grid-template:
 - grid
- Espaçamento
 - column-gap
 - row-gap
 - gap
- Alinhamento de Itens
 - justify-items: **start** | **end** | **center** | **stretch**
 - align-items: **start** | **end** | **center** | **stretch**
 - place-items: [**justify-items**] / [**align-items**]



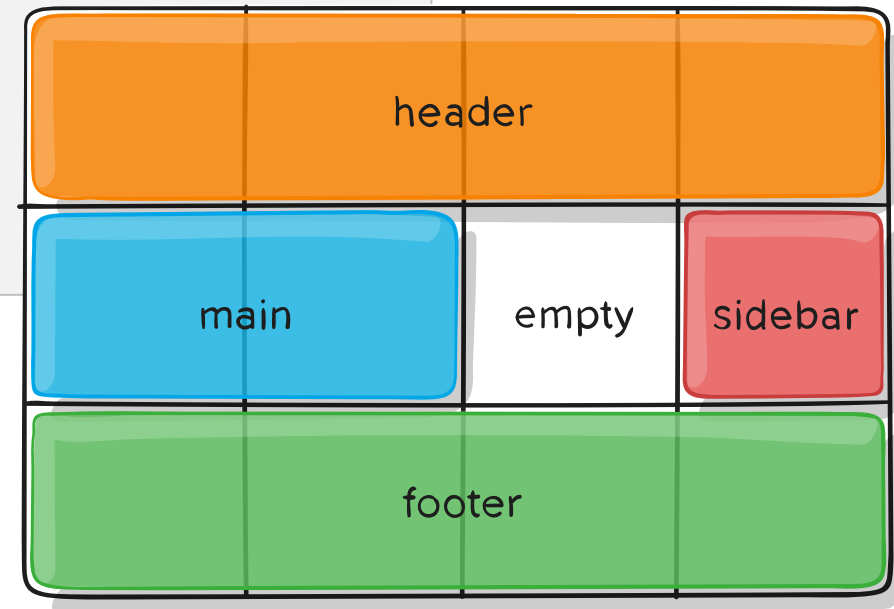
Layout Grid

* CSS-TRICKS

Propriedades de Containers – Exemplos – Definição de Colunas e Áreas



```
.container {  
  display: grid;  
  grid-template-columns: 50px 50px 50px 50px;  
  grid-template-rows: auto;  
  grid-template-areas:  
    "header header header header"  
    "main main . sidebar"  
    "footer footer footer footer";  
}
```



Layout Grid

* CSS-TRICKS

Propriedades de Items

- Localização

- grid-row-start: <number> | <name> | span <number> | span <name> | auto
- grid-row-end: <number> | <name> | span <number> | span <name> | auto
- grid-row: <start-line> / <end-line> | <start-line> / span <value>
- grid-column-start: <number> | <name> | span <number> | span <name> | auto
- grid-column-end: <number> | <name> | span <number> | span <name> | auto
- grid-column: <start-line> / <end-line> | <start-line> / span <value>
- grid-area: <name> | <row-start> / <column-start> / <row-end> / <column-end>

- Alinhamento

- justify-self: start | end | center | stretch
- align-self: start | end | center | stretch
- place-self: [justify-self] / [align-self]

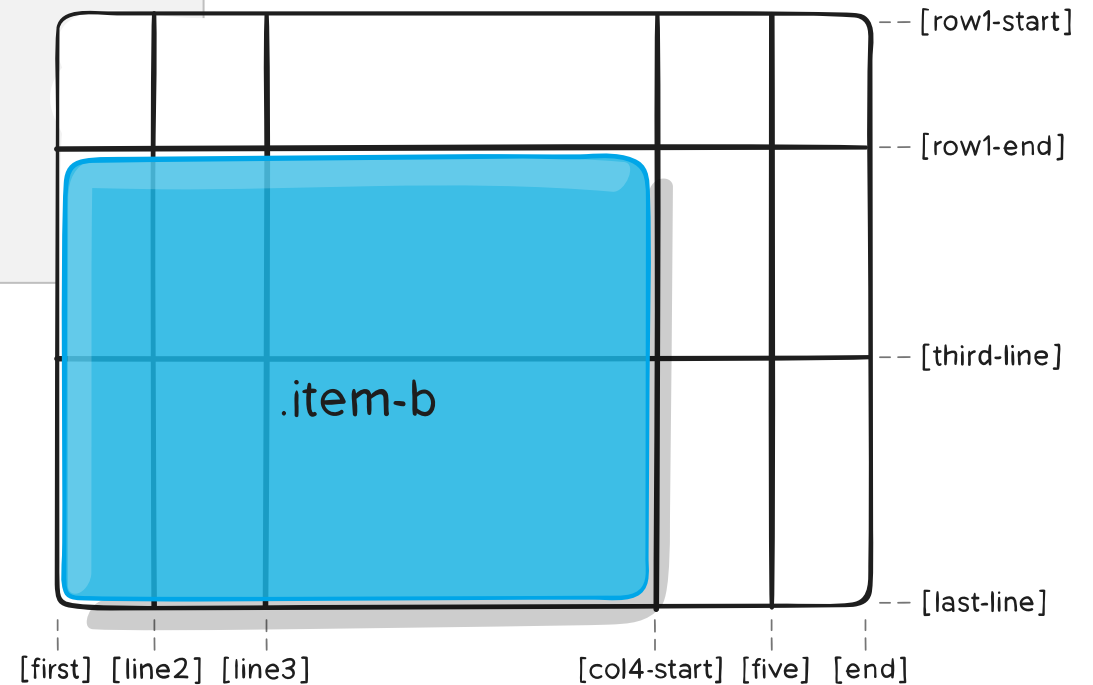
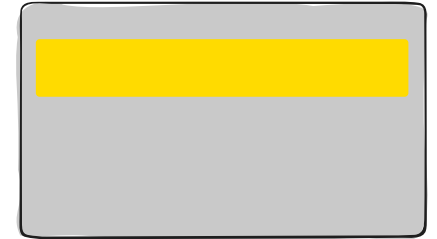


Layout Grid

* CSS-TRICKS

Propriedades de Items – Exemplos - Posicionamento de item

```
.item-b {  
  grid-column-start: 1;  
  grid-column-end: span col4-start;  
  grid-row-start: 2;  
  grid-row-end: span 2  
}
```

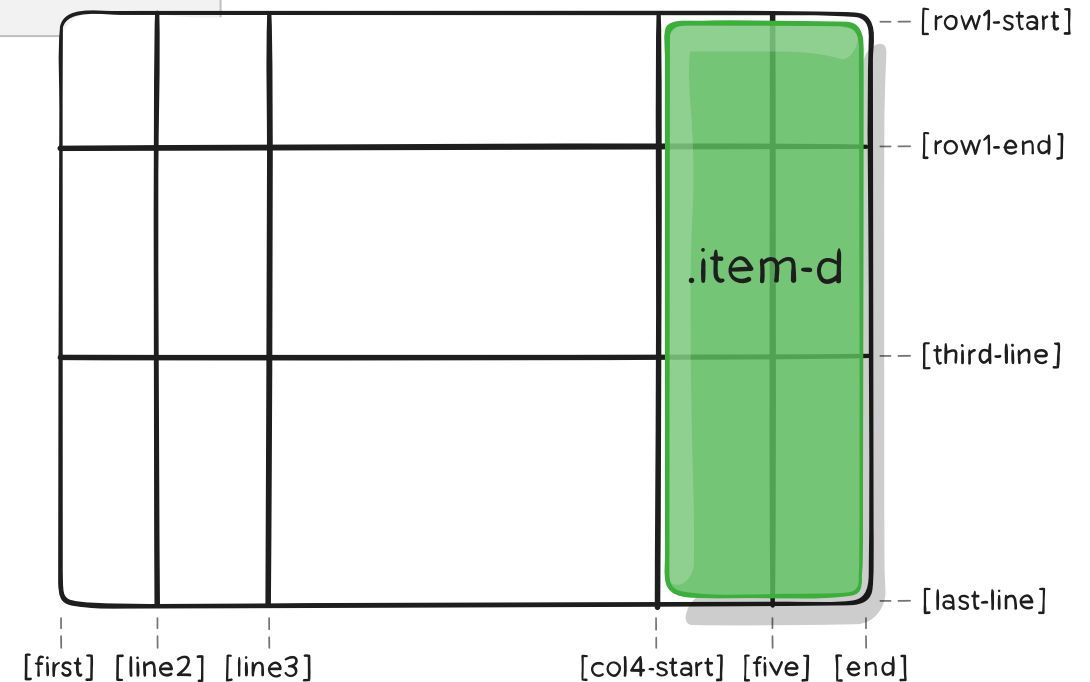
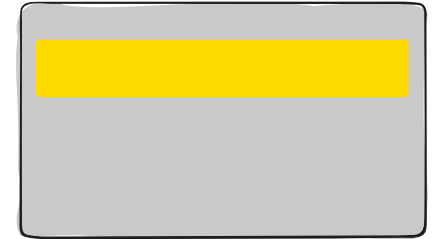


Layout Grid

* CSS-TRICKS

Propriedades de Items – Exemplos - Posicionamento de item

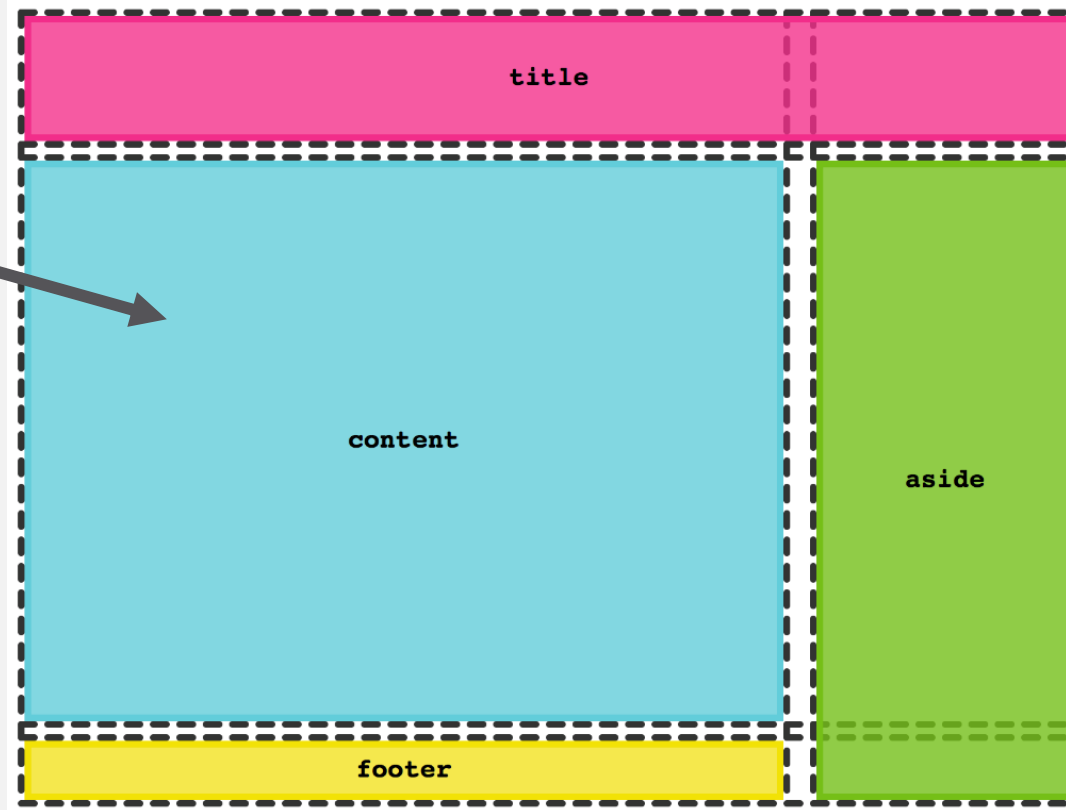
```
.item-d {  
  grid-area: 1 / col4-start / last-line / 6;  
}
```



Layout Grid – Exemplo

```
#wrapper {  
  display: grid;  
  grid-gap: 10px 20px;  
  grid-template-rows: 100px 1fr auto;  
  grid-template-columns: 1fr 200px;  
  grid-template-areas:  
    "header header"  
    "content aside "  
    "footer aside ";  
}
```

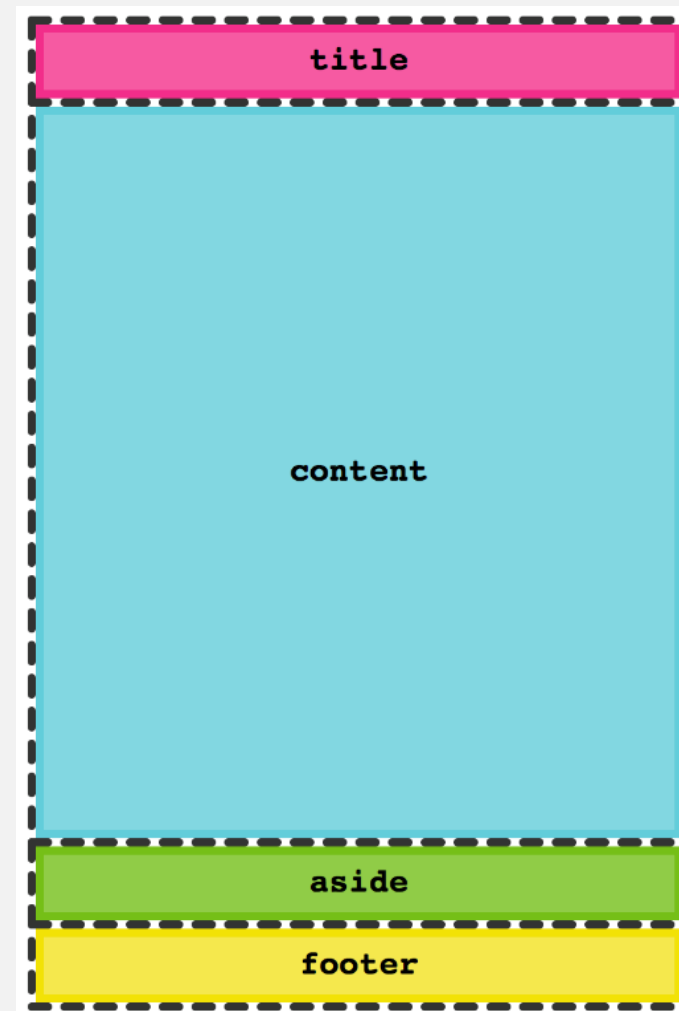
```
@media (max-width: 600px) {  
  #wrapper {  
    grid-gap: 0;  
    grid-template-rows:  
      auto 1fr auto auto;  
    grid-template-columns: 1fr;  
    grid-template-areas:  
      "header "  
      "content"  
      "aside "  
      "footer ";  
  }  
}
```



Layout Grid – Exemplo

```
#wrapper {  
  display: grid;  
  grid-gap: 10px 20px;  
  grid-template-rows: 100px 1fr auto;  
  grid-template-columns: 1fr 200px;  
  grid-template-areas:  
    "header header"  
    "content aside "  
    "footer aside ";  
}
```

```
@media (max-width: 600px) {  
  #wrapper {  
    grid-gap: 0;  
    grid-template-rows:  
      auto 1fr auto auto;  
    grid-template-columns: 1fr;  
    grid-template-areas:  
      "header "  
      "content"  
      "aside "  
      "footer ";  
  }  
}
```



Obrigado!